

CHAPTER 15

Probabilistic Models

15.1. Introduction

Chapter 14 provided background into some of the deterministic areas of operations research, those primarily having to do with mathematical programming and search procedures. Very often, however, when we set out to develop a mathematical model of a real-world system, we find that certain important elements are random and we cannot satisfactorily ignore their variation. We then turn to probabilistic operations research models to help us quantify, evaluate, and optimize the system. This chapter includes three areas of probabilistic models including queueing, inventory, and Markov chains. Chapter 16 deals with simulation, which is applicable for use in virtually all areas of probabilistic modeling. The intent, then, of this chapter is to introduce probabilistic O.R. modeling, analysis, and design.

15.2. Queueing Theory

Queueing theory is the study of waiting lines. A waiting line, or queue, is a common occurrence whenever the service facility cannot keep pace with the demand for the service. We are involved in queueing almost every day of our lives. For example, waiting in a grocery checkout line, buying books at the beginning of a school term,

bank
→
registrars
office
→ ER's
in hospital

or even getting a drink of water from a water fountain constitute queueing situations. In addition, organizations providing services are often called upon to make queueing-related decisions. The number of tool crib clerks to use in a large machine shop, the number of turnpike toll-takers, or the capacity of waiting space for in-process inventory items are examples.

Queueing theory is particularly interesting, for it is almost exclusively probabilistic. That is, no one is capable of predicting exactly when arrivals for service will take place and no one can predict exactly how long a service will take. Scheduling and appointments are often used to smooth the arrival pattern, for example, as in a doctor's office. Arrivals are often either early or late, however, and servers require variable amounts of time, resulting again in a probabilistic situation.

Queueing theory enables us to use mathematics to describe and make decisions about waiting lines under many different conditions. As an example, we could always please customers by providing an excess of servers in order to minimize delay or waiting time. Obviously, this would be costly. We could also provide minimal service and just let the customers wait in line. This, too, would be costly because customers would balk (not enter the line) or decide not to return in the future. We, therefore, seek a balance between these two extremes. Queueing theory provides us information for properly designing our service facility.

15.2.1 Queueing System Structure

A basic queueing system is made up of a waiting line and a service facility. Arrivals to the queueing system come from the calling population. These elements are illustrated in Figure 15.1. Each of these elements has characteristics that define the queueing process.

The population, for example, can be infinite (practically speaking, very large) or finite. It is characterized by an arrival rate (λ) and an arrival distribution. It is much easier to model an infinite population instead of one that is finite, because with a finite population the arrival rate is affected by the number of units already in the queueing system. The arrival distribution can vary widely, but it is commonly assumed to be Poisson (i.e., the number of units arriving for service in an interval of time is Poisson distributed).

The queue may also be infinite or of limited size. For example, almost all ticket queues can grow as large as may ever be required. A "do-it-yourself" spray car wash, however, has limited waiting room for drivers waiting their turns. Again, the infinite queue is easier to model mathematically. The service discipline relates to the way in which units leave the queue for service. The order may be first-come-first-served, last-come-first-served, random, based on priorities, and so on. We usually assume first-come-first-served unless otherwise specified.

The service facility is characterized by the number and configuration of servers. There may be only one server or there may be several servers in parallel. If the service requires several steps, there may be servers in series. The service facility is also characterized by a rate of service (μ) and a distribution of service

→ # of service providers
→ capacity of waiting room

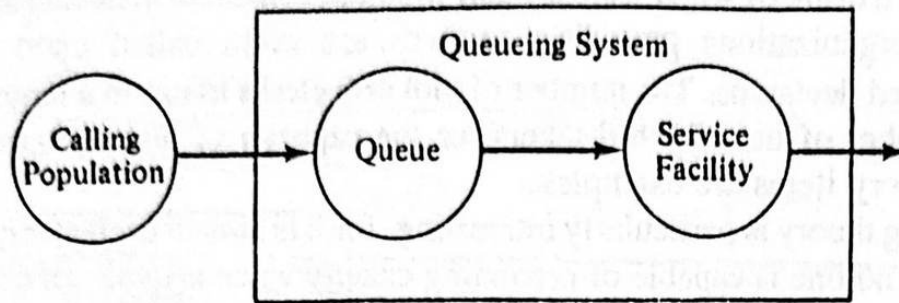


Fig. 15.1. Basic queueing system.

times. Frequently used service time distributions include the gamma (not treated in Appendix A), the exponential, and a constant service time.

15.2.2 Queueing Notation

In the remaining sections on queueing some of the basic, yet most frequently used, models are presented. The notation used in the equations may be summarized as follows:

$P(n)$ = probability that exactly n population units are in the queueing system, including both waiting line and service facility

λ = arrival rate in units per time

μ = service rate in units per time

N = expected number of units in the queueing system, including both waiting line and service facility

N_q = expected number of units in the waiting line

T = expected time in the queueing system, including both waiting line and service facility

T_q = expected time in the waiting line

In each of the following models there are two relationships between expected numbers of units and waiting times. Those relationships are as follows:

$$N = \lambda T \quad \text{Little's Law} \quad (15.1)$$

and

$$N_q = \lambda T_q \quad (15.2)$$

Thus, if we calculate the expected number of units, it is easy to get expected time, or vice versa. Also, a relationship between waiting time in the system and waiting time in the queue is as follows:

$$T = T_q + \frac{1}{\mu} \quad (15.3)$$

This equation states that the expected time in the queueing system equals the expected time in the waiting line plus the mean service time.

15.2.3 Single-Service Channel

Three queueing models will be presented in this section. Each is for a service facility having but one server, an infinite population, and an infinite queue. We shall assume that the number of arrivals occurring over intervals of time are Poisson distributed. The distribution of service times, however, will be exponential, arbitrary, and constant, respectively, in the three sections.

Exponential Service Times. With Poisson arrivals and exponential service times, the probability of no units in the queueing system is

$$P(0) = 1 - \frac{\lambda}{\mu} \quad (15.4)$$

The probability of having a number n units in the system is

$$P(n) = P(0) \left(\frac{\lambda}{\mu} \right)^n \quad (15.5)$$

The expected (average) number of units in the queueing system is

$$N = \frac{\lambda}{\mu - \lambda} \quad (15.6)$$

and the expected time in the system is

$$T = \frac{1}{\mu - \lambda} \quad (15.7)$$

Similarly, the expected number of units in the queue is

$$N_q = \frac{\lambda^2}{\mu(\mu - \lambda)} \quad (15.8)$$

and the expected waiting time there is

$$T_q = \frac{\lambda}{\mu(\mu - \lambda)} \quad (15.9)$$

As an example, suppose that the number of arrivals is Poisson distributed with a mean of 15 per hour. Service times are exponentially distributed with a mean of 3 minutes each (20 services per hour). The probability that no units are in the system is

$$P(0) = 1 - \frac{\lambda}{\mu} = 1 - \frac{15}{20} = 0.25$$

The probability that three units are in the system is

$$P(3) = P_0 \left(\frac{\lambda}{\mu} \right)^3 = 0.25 \left(\frac{15}{20} \right)^3 = 0.1055$$

The other queueing system characteristics of interest include the following:

$$N = \frac{\lambda}{\mu - \lambda} = \frac{15}{20 - 15} = 3 \text{ persons in the system}$$

$$T = \frac{1}{\mu - \lambda} = \frac{1}{20 - 15} = 0.2 \text{ hour in the system}$$

$$N_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{15^2}{20(20 - 15)} = 2.25 \text{ persons in the queue}$$

$$T_q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{15}{20(20 - 15)} = 0.15 \text{ hour in the queue}$$

Also, the server is utilized a fraction ρ of the time where

$$\rho = \frac{\lambda}{\mu} = \frac{15}{20} = 0.75$$

or 75% of the time.

Arbitrary Service Times. For arbitrary service times, the mean of the service time distribution, $1/\mu$, and its variance, σ^2 , must be known. The actual distribution, however, does not need to be known or stated. The following results apply when $\lambda/\mu < 1$:

$$N = \frac{\lambda^2 \sigma^2 + \left(\frac{\lambda}{\mu}\right)^2}{2 \left(1 - \frac{\lambda}{\mu}\right)} + \frac{\lambda}{\mu} \quad (15.10)$$

$$T = \frac{N}{\lambda} = \frac{\lambda \sigma^2 + \frac{\lambda}{\mu^2}}{2 \left(1 - \frac{\lambda}{\mu}\right)} + \frac{1}{\mu} \quad (15.11)$$

$$N_q = N - \frac{\lambda}{\mu} = \frac{\lambda^2 \sigma^2 + \left(\frac{\lambda}{\mu}\right)^2}{2 \left(1 - \frac{\lambda}{\mu}\right)} \quad (15.12)$$

$$T_q = \frac{N_q}{\lambda} = \frac{\lambda \sigma^2 + \frac{\lambda}{\mu^2}}{2 \left(1 - \frac{\lambda}{\mu}\right)} \quad (15.13)$$

Again consider the previous example where arrivals are Poisson with $\lambda = 15$. Let service times be of unknown distribution, but have a mean of 3 minutes each (20 services per hour) and a standard deviation (σ) of 5 minutes (0.083333 hours) and hence $\sigma^2 = 0.006944$. Now

$$N = \frac{\lambda^2 \sigma^2 + \left(\frac{\lambda}{\mu}\right)^2}{2 \left(1 - \frac{\lambda}{\mu}\right)} + \frac{\lambda}{\mu} = \frac{15^2 (0.006944) + \left(\frac{15}{20}\right)^2}{2 \left(1 - \frac{15}{20}\right)} + \frac{15}{20} = 5.0 \text{ persons}$$

and values of $T = 0.333$ hour, $N_q = 4.25$ persons, and $T_q = 0.283$ hour follow easily.

We should notice that when the service time distribution is exponential with mean $1/\mu$, and, therefore, with variance $\sigma^2 = 1/\mu^2$, our arbitrary service time distribution results revert to the exponential results in the previous section. Also, we can see that N , T , N_q , and T_q all increase as σ^2 increases. Thus, the variability of the server has an important effect on the performance measures of the queueing system.

Constant Service Times. Many manufacturing systems involve automated processing of certain operations. This automated processing often takes a constant amount of time to perform. In this case, arrivals to the work station may still be Poisson distributed, but service times will be of duration $1/\mu$, with no time dispersion, $\sigma^2 = 0$. In this case, our equations from the preceding section reduce to

$$N = \frac{\left(\frac{\lambda}{\mu}\right)^2}{2 \left(1 - \frac{\lambda}{\mu}\right)} + \frac{\lambda}{\mu} \quad (15.14)$$

$$T = \frac{\frac{\lambda}{\mu^2}}{2 \left(1 - \frac{\lambda}{\mu}\right)} + \frac{1}{\mu} \quad (15.15)$$

$$N_q = \frac{\left(\frac{\lambda}{\mu}\right)^2}{2 \left(1 - \frac{\lambda}{\mu}\right)} \quad (15.16)$$

$$T_q = \frac{\frac{\lambda}{\mu^2}}{2 \left(1 - \frac{\lambda}{\mu}\right)} \quad (15.17)$$

Continuing with our previous example having Poisson arrivals with $\lambda = 15$, but now with constant service at a rate of $\mu = 20$, our performance measures are: $N = 1.875$ persons, $T = 0.125$ hour, $N_q = 1.125$ persons, and $T_q = 0.075$ hour.

15.3. Inventory Control

In Chapter 7 you were introduced to the concept of inventory control, as well as a discussion of the relevant considerations and costs involved. Recall that there is much emphasis and interest today in minimizing inventories and storage by proper sizing of order quantities and maximizing turnover of stock. Inventory models help us determine the costs and consequences of various inventory levels. Some of the models presented in Chapter 7 were deterministic in nature. The demand for an item over a fixed interval of time or the lead time from order to receipt is not always constant, however. That is, demand may vary from day to day in accordance with a probability distribution. Similarly, lead time may be either short or long. If this variability is significant, it may be necessary for us to explicitly take the stochastic nature of demand and/or lead time into account in developing our mathematical model.

Three different inventory models will be discussed in the following sections. The first covers a single period with no setup cost. This is, perhaps, the simplest model that we can use to illustrate some of the considerations required in developing a probabilistic inventory model. The following two sections are devoted to the lot size-reorder point and periodic review models. For these we present only philosophy because of the difficulty of their development.

15.3.1 Single-Period Model—No Setup Cost

A classic problem often used to illustrate the nature of probabilistic inventory models is the single-period model with no setup cost. Units are purchased at a cost of C dollars per item to be sold during a single period for S dollars per item. The salvage value of any leftover items at the end of the period is equal to M dollars per unit. The quantity purchased at the beginning of the period is denoted by Q , and the number of units sold during the period is given by X . Of course, X represents demand for the items and is a random variable described by the distribution $p(X)$.

This model is applicable to the inventory control of an item that becomes obsolete at the end of the period. For example, bakery goods such as bread and donuts have their primary demand shortly after being made. They also have a "salvage" market at a reduced price if some remain unsold after the first day.

Christmas tree inventories are well described by this model, as are Valentine candy and seasonal greeting cards. This has also been called the "newsboy" model, because it applies to the inventory of newspapers that should be maintained by a newsstand on any given day. In each of these cases, the objective is to purchase the optimum quantity of items at the beginning of the period in order to maximize profits over the period.

Let's take a simple example and assume that we belong to a charitable organization which decides to operate a Christmas tree stand to raise money for a worthy cause. We must place an order with a local Christmas tree supplier a month in advance for the number of trees we want to purchase at \$10 each. We can then sell the trees individually for \$18 each. Any trees left unsold will be removed after Christmas at a cost of \$1 each. Note that this represents a negative "salvage value," or $M = -1$. We estimate that from 201 to 300 trees can be sold, with the probability of any number in this range being $p(X) = 0.01$. That is, we believe that a uniform distribution will apply with the probability of selling 201, 202, . . . , 300 trees each being $1/100$.¹

The number we order, Q , should be chosen carefully because for each tree demanded when we are out of stock, a profit of \$8 is lost. For each tree left over, however, we have lost not only the \$10 originally paid, but also \$1.00 for removal. To better define the problem, we should explicitly state the costs and revenues involved as a function of demand, X . This is accomplished in Table 15.1.

It would be a simple matter to decide on a value of the order quantity, Q , and then for a particular demand, X , calculate what the profit would be. Unfortunately, that will not be of much use to us. Instead, since demand is unknown, we must write an expression for the *expected value* of profit for a given order quantity and then find the order quantity that maximizes the expected value.

We recall that the expected value of a random variable X having distribution $p(X)$ is written

$$E(X) = \sum_X Xp(X)$$

Similarly, we can write an expression for the expected value of a function of X , say $g(X)$, as

$$E[g(X)] = \sum_X g(X)p(X)$$

Since profit is a function of demand, we have

$$\begin{aligned} E(\text{profit}) = E[g(X)] &= \sum_{X=0}^Q [SX + M(Q - X) - CQ] p(X) \\ &+ \sum_{X=Q+1}^{\infty} (SQ - CQ) p(X) \end{aligned} \quad (15.18)$$

¹This distribution is chosen here for its simplicity rather than for its reality.

Table 15.1. SINGLE-PERIOD INVENTORY COSTS AND REVENUES

Demand X	Revenue	Cost	Profit $g(X)$
$X \leq Q$	$= SX + M(Q - X)$ $18X - (Q - X)$	CQ $10Q$	$SX + M(Q - X) - CQ$ $18X - (Q - X) - 10Q$
$X > Q$	$= SQ$ $18Q$	CQ $10Q$	$SQ - CQ$ $18Q - 10Q$

In our example problem, since $p(X)$ is defined only over $X = 201, \dots, 300$, expected profit may be written as follows:

$$E(\text{profit}) = E[g(X)] = \sum_{X=201}^Q [19X - 11Q] p(X) + \sum_{X=Q+1}^{300} 8Q p(X)$$

One way to optimize the order quantity is to try different values of Q in the expression above. For example, let us see whether we prefer to order $Q = 250$ or $Q = 275$ trees.

At $Q = 250$:

$$E(\text{profit}) = \sum_{X=201}^{250} [19X - 11(250)] p(X) + \sum_{X=251}^{300} 8(250) p(X) = \$1,767.25$$

At $Q = 275$:

$$E(\text{profit}) = \sum_{X=201}^{275} [19X - 11(275)] p(X) + \sum_{X=276}^{300} 8(275) p(X) = \$1,672.75$$

Obviously, we would rather order 250 trees than 275, but is 250 the true optimum? Continuing as we have above, we would try to find the value of Q which gives a higher expected profit than either $Q - 1$ or $Q + 1$.

Alternatively, we could difference² (Equation 15.18) and determine that the optimum order quantity is the largest Q for which

$$1 - F(Q - 1) \geq \frac{C - M}{S - M} \quad (15.19)$$

where $F(Q - 1)$ is the cumulative distribution function, or $\sum_{X=0}^{Q-1} p(X)$. For our example, this would be

$$1 - F(Q - 1) \geq \frac{10 - (-1)}{18 - (-1)} = \frac{11}{19} = 0.58$$

and the optimum order quantity is $Q_0 = 243$.

²Differencing a discrete function is akin to differentiating a continuous function.

Sure enough, if we calculate the expected value of profit for $Q = 242, 243$, and 244 , we obtain \$1,772.41, \$1,772.43, and \$1,772.26, respectively.

15.3.2 Lot Size—Reorder Point Models

In the lot size–reorder point model an order for Q items is placed when the inventory level drops to a *reorder point*, r . This, of course, requires a data base in which we know the quantity of items remaining on-hand after each transaction. We saw an example of such a model in Figure 14.3 and an extension of that model is considered here.

The example shown in Figure 14.3 had a stockout condition resulting from probabilistic lead time. If we do not wish stockouts, we can reorder earlier than that given by the deterministic model by adding a quantity called safety stock (SS) to the reorder quantity (RQ) and reorder when the stock on hand reaches RQ' where

$$RQ' = RQ + SS$$

Figure 15.2 shows what might happen in a situation like that in Figure 14.3 when safety stock is used. Notice that the shape of this graph is identical to that of Figure 14.3 except that it has been shifted up by an amount equal to the safety stock, so that stockouts do not occur as often (zero times in this case).

The amount of safety stock necessary can be determined in two ways. First, we can experiment with various levels of safety stock until we get the performance we desire, or we can determine the amount statistically. To determine the amount statistically, we only need to know the distribution of the lead time, its mean and standard deviation, and the service level desired where service level is defined as the percentage of time we will allow stockouts.

For example, suppose that the lead time is normally distributed with a mean of 20 days and a standard deviation of 2 days. Suppose further that management wants us to be sure to have stock available 98% of the time (stated another way, stockouts are allowed only 2% of the time). We can then go to a cumulative standard normal table and find the point that corresponds to 98%.³ In the normal table of Appendix B we find that this corresponds to approximately $+2.05\sigma$. The reorder point then should be based on demand for a lead time of

$$20 + 2.05(2) = 24.1 \text{ days}$$

If demand is 10 items per day, the reorder point is

$$(24.1 \text{ days})(10 \text{ items/day}) = 241 \text{ units}$$

Another illustration of a lot size–reorder point model is shown in Figure 15.3. Note that the time from placement of the order to receipt of the order is again a random variable. Also, the wavy line indicates that demand is now also a random variable. Under these stochastic conditions, we see that sometimes there is stock

³ Assuming that the demand rate is constant and lead time is normally distributed.

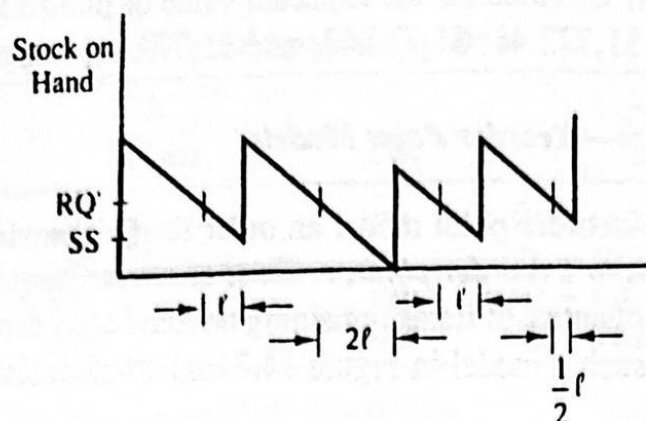


Fig. 15.2. Probabilistic lead time with safety stock.

on hand when a new order is received. At other times, stockouts and back orders or lost sales are incurred.

Expressing this model mathematically is difficult. It is comprised of a setup charge for placing the order (neglected on the single-period model in the previous section), an inventory carrying charge for stock on hand, and a good will or profit loss charge for shortages. The demand and lead time distributions are explicitly included in the model and conventional optimization methods are used to solve for Q and r —the optimum lot size and reorder point.⁴

15.3.3 Periodic Review Models

Rather than have the timing of orders be dictated by on-hand inventory reaching a reorder point, we can employ the *periodic review* strategy. This philosophy requires that we review a particular stock at intervals of time, T . If any units at all have been demanded since the last review, an order is placed. This time, however, the order is sized to raise our “on-hand *plus* on-order” inventory level to an amount R . This philosophy is illustrated in Figure 15.4.

We can see that the order quantity varies in this model, depending on the inventory level at the time our order is placed. Also, although on-hand inventory is not shown, we can have units on hand or shortages, depending on lead times and demands. Once again, our mathematical model reflects setup, inventory carrying, and shortage costs, which we seek to minimize by properly selecting the level R and the periodic review interval T .⁵

⁴A good discussion of this model is presented in G. Hadley and T. M. Whitin, *Analysis of Inventory Systems* (Englewood Cliffs, N.J.: Prentice Hall, 1963), pp. 159–175.

⁵This model is also discussed in Hadley and Whitin.

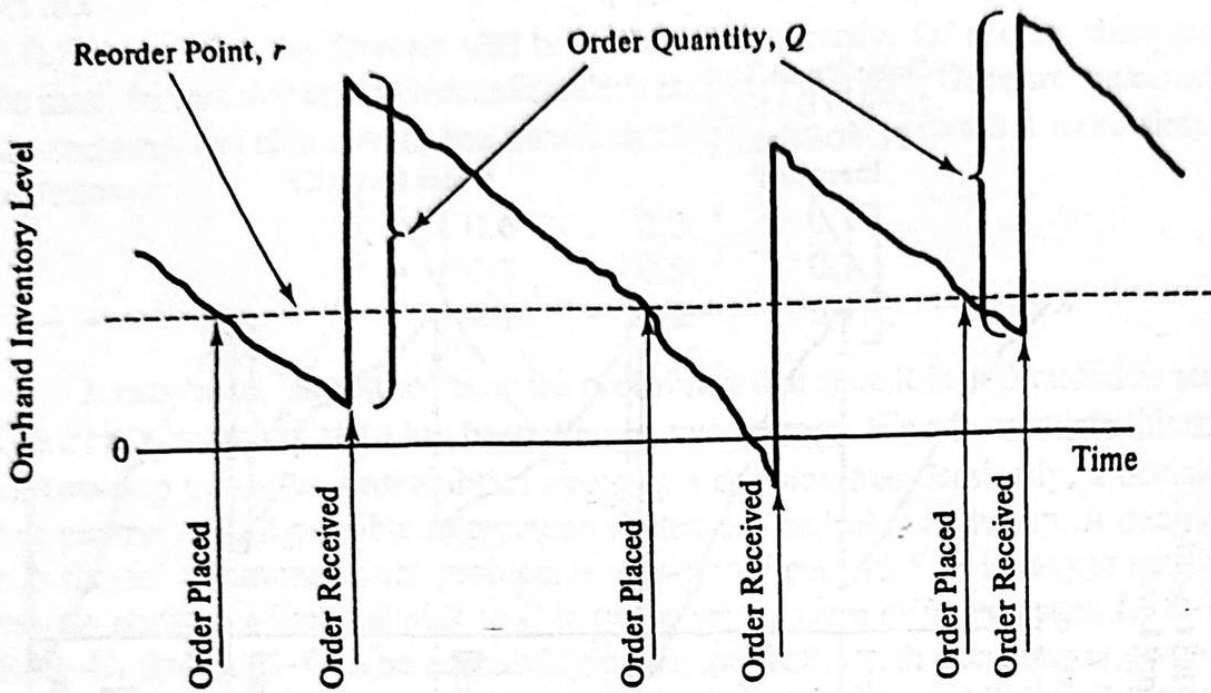


Fig. 15.3. Lot size-reorder point model.

15.4. Markov Chains

A *Markov chain* represents the behavior of a stochastic process⁶ that can be in one and only one of M states, $1, 2, \dots, M$, at time t , $t = 1, 2, \dots$. The process is free to switch from one state to another at the next point in time. Such a change, say from state i to state j , between two consecutive points in time is done with *transition probability* p_{ij} .

We can represent a Markov chain as follows:

		To state			
		1	2	$\dots$	M
From state	1	p_{11}	p_{12}	$\dots$	p_{1M}
	2	p_{21}	p_{22}	$\dots$	p_{2M}
	$\cdot$	$\cdot$	$\cdot$		$\cdot$
	$\cdot$	$\cdot$	$\cdot$		$\cdot$
	M	p_{M1}	p_{M2}	$\dots$	p_{MM}

We should note several important facts about this chain. First, there is only a finite number of states. Second, the probability of making a transition from one state to

⁶A sequence of numbers representing inventory levels or inventory demands over consecutive days or weeks would be considered a stochastic process. Similarly, the numbers of arrivals to or services from a queueing system per unit time represent a stochastic process.

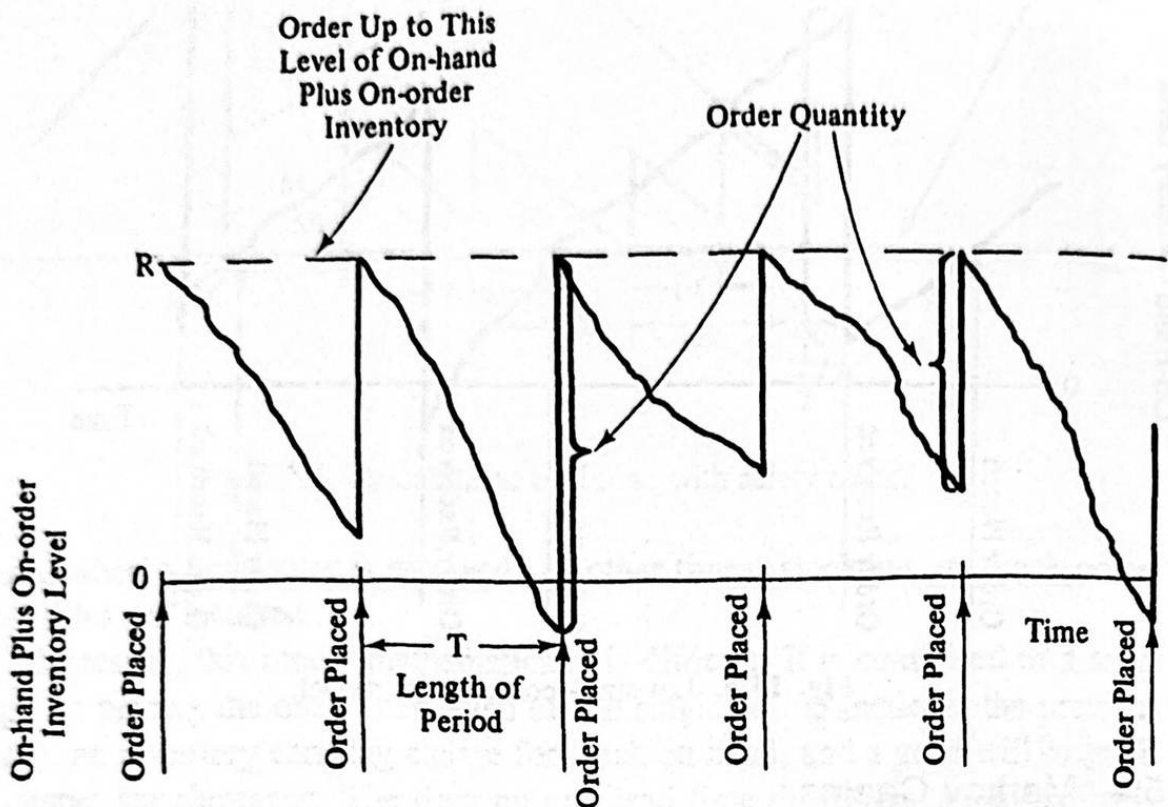


Fig. 15.4. Periodic review model.

another depends only on the present state occupied, not on any earlier history. Third, the transition probabilities p_{ij} are stationary and do not change over time.⁷

15.4.1 Regular Markov Chains

Let us now represent a communications problem as a Markov chain. Suppose that a weather forecast is passed along from person to person. The forecast has three possible states: sleeting (S), rainy (R), and cloudy (C).⁸ The transition probabilities for two successive persons are given in the Markov chain transition matrix presented as follows:

		To state		
		S	R	C
From state	S	0.6	0.3	0.1
	R	0.2	0.5	0.3
	C	0.1	0.2	0.7

We can see that if one person understands the forecast to be, say, rainy, there is a 0.2 chance that the next person told will understand the forecast to be sleeting. Or, there is a 0.3 probability that the forecast will be understood as cloudy. This leaves

⁷Markov chains receive an elementary treatment in John G. Kemeny et al., *Finite Mathematical Structures* (Englewood Cliffs, N.J.: Prentice Hall, 1959).

⁸The number of states is held to three to permit ease of presentation. More realistically, we should have several additional weather states or, alternatively, one additional state such as "other."

a 0.5 chance that the forecast will be understood correctly. Of course, there may be many factors that could cause misunderstanding of this sort. Once we understand the meaning and elements of the transition matrix, we can express it more simply as follows:

$$P = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.2 & 0.7 \end{bmatrix}$$

It may be of interest to know the probability that state R is understood as state C after the communication has been through two persons. We can calculate this and all two-step transition probabilities by using a decision tree. Basically, a decision tree enumerates all possible interpretations that can be had at each step. A decision tree for our communications problem is shown in Figure 15.5. It is easy to see that we can go from a forecast of R to C in two steps via three different paths, R-S-C, R-R-C, and R-C-C. The probability of any particular path occurring is given at the end of each path and represents the product of the probabilities of the two transitions along the path. The overall probability that we go from state R to C in two steps is the sum of the probabilities of each path ending in C, or $0.02 + 0.15 + 0.21 = 0.38$.

An easier way to determine the two-step transition probabilities is to calculate the two-step transition matrix. This is done through the matrix multiplication of P times P . For example, P^2 is calculated as follows:

$$\begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.2 & 0.7 \end{bmatrix} \times \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.2 & 0.7 \end{bmatrix} = \begin{bmatrix} 0.43 & 0.35 & 0.22 \\ 0.25 & 0.37 & 0.38 \\ 0.17 & 0.27 & 0.56 \end{bmatrix}$$

Now, we can read the two-step transition probability from a forecast of R to one of C directly as .38. In addition, though, we have the two-step transition probabilities from each state to every other state.

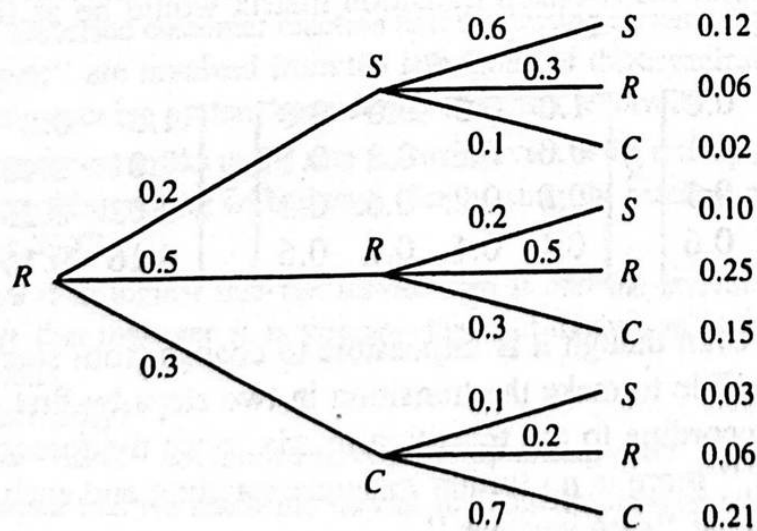


Fig. 15.5. Communications decision tree.

In general, we can calculate n -step transition matrices by using

$$P \cdot P^{n-1} = P^{n-1} \cdot P = P^n \quad (15.20)$$

If we were to calculate P^n for a large value of n in our communications example, the result would be

$$P^n = \begin{bmatrix} 0.265 & 0.324 & 0.412 \\ 0.265 & 0.324 & 0.412 \\ 0.265 & 0.324 & 0.412 \end{bmatrix} \quad (\text{for large } n)$$

This result is interesting because it points out the obvious: After our forecast has been passed along many times, the probability that the n th person understands it as sleeting, rainy, or cloudy is independent of the actual original forecast.

15.4.2 Absorbing Markov Chains

Often in modeling with Markov chains we have one or more absorbing states. When the process makes a transition to an absorbing state, it is impossible to leave that state. For example, let us suppose that another state is added to our communications problem. That state can be labeled "nice weather," or N, and it fits into a one-step transition matrix as follows:

$$\text{From state} \quad \begin{array}{c|cccc} & \text{N} & \text{S} & \text{R} & \text{C} \\ \hline \text{N} & 1 & 0 & 0 & 0 \\ \text{S} & 0 & 0.6 & 0.3 & 0.1 \\ \text{R} & 0 & 0.2 & 0.5 & 0.3 \\ \text{C} & 0.1 & 0.1 & 0.2 & 0.6 \end{array}$$

Again, we can calculate the two-, three-, and n -step transition probabilities as before. For example, the two-step transition matrix would be as follows:

$$\begin{array}{c} P \\ \begin{bmatrix} 1.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.6 & 0.3 & 0.1 \\ 0.0 & 0.2 & 0.5 & 0.3 \\ 0.1 & 0.1 & 0.2 & 0.6 \end{bmatrix} \end{array} \times \begin{array}{c} P \\ \begin{bmatrix} 1.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.6 & 0.3 & 0.1 \\ 0.0 & 0.2 & 0.5 & 0.3 \\ 0.1 & 0.1 & 0.2 & 0.6 \end{bmatrix} \end{array} = \begin{array}{c} P^2 \\ \begin{bmatrix} 1.0 & 0.0 & 0.0 & 0.0 \\ 0.01 & 0.43 & 0.35 & 0.21 \\ 0.03 & 0.25 & 0.37 & 0.35 \\ 0.16 & 0.16 & 0.25 & 0.43 \end{bmatrix} \end{array}$$

Here we see that even though it is impossible to change from state S or R to N in one step, it is possible to make the transition in two steps by first going to state C and then to N. According to our transition matrix, once the forecast is understood as "nice weather," there is no further misunderstanding and each communication conveys a forecast of "nice weather."

In the long run, the P^n matrix for large n is given by

$$P^n = \begin{bmatrix} 1.0 & 0.0 & 0.0 & 0.0 \\ 1.0 & 0.0 & 0.0 & 0.0 \\ 1.0 & 0.0 & 0.0 & 0.0 \\ 1.0 & 0.0 & 0.0 & 0.0 \end{bmatrix}$$

This verifies our intuition that, after passing through *many* intermediaries, our forecast will be understood as “nice weather” regardless of the original forecast.

15.5. Impact of Statistics and Computers

Nearly all industrial engineering programs require a strong statistics component in which variation is studied. This is important, since virtually everything in engineering (and even life in general) varies. The engineer who has a background in probability and statistics is far out ahead of the pack.

All probabilistic models make an attempt to model a real-world situation more accurately by explicitly addressing the inherent variation. To develop, analyze, or optimize the model, however, requires a background in probability and/or statistics. The added complexity also most often requires use of the computer for any except the most simplified problems. Fortunately, there are personal computer probabilistic O.R. software packages that provide help. Of course, the old standby, spreadsheets, can be used to do a great deal of the work on many small but realistic problems. Finally, when the probabilistic problem becomes so complex that it is not easily modeled analytically, we turn to simulation, the subject of Chapter 16.